

JAMES RUSE AGRICULTURAL HIGH SCHOOL
TERM 1 ASSESSMENT 1998
YEAR 12 3/4 UNIT

Time allowed: 85 minutes.

All questions are to be attempted

Each question is to be handed in separately.

Question 1 : START A NEW PAGE

(a) Evaluate exactly : $\sin^{-1} \left(\frac{-1}{2} \right)$

(b) Evaluate : $\lim_{x \rightarrow 0} \left(\frac{\sin 4x}{2x} \right)$

(c) Differentiate with respect to x :

(i) $y = 2 \tan (x^3)$

(ii) $y = x \sin^{-1} x$

(d) Find the exact value of the gradient to the curve $y = \operatorname{cosec} x$ at $x = \frac{7\pi}{6}$.

(e) Find $\frac{d}{dx} \cos^{-1} \left(\frac{1}{x} \right)$ in simplest terms.

Question 2 : START A NEW PAGE

(a) Evaluate : $\int_{-4}^4 \frac{6 dx}{x^2 + 16}$

(b) Find : $\int \frac{2 dx}{\sqrt{3 - 4x^2}}$

(c) Find : $\int \frac{4x + 5 dx}{\sqrt{9 - x^2}}$

(d) (i) Express $143 \cos \theta - 24 \sin \theta$ in the form $R \cos (\theta + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give α correct to four decimal places.

(ii) Solve the equation $143 \cos \theta - 24 \sin \theta = 100$, for $0 < \theta < 2\pi$, giving answer correct to four decimal places.

Question 3 : START A NEW PAGE

(a) Find the sum of : $3^n + 3^{n-1} + \dots + 3^{-2n}$

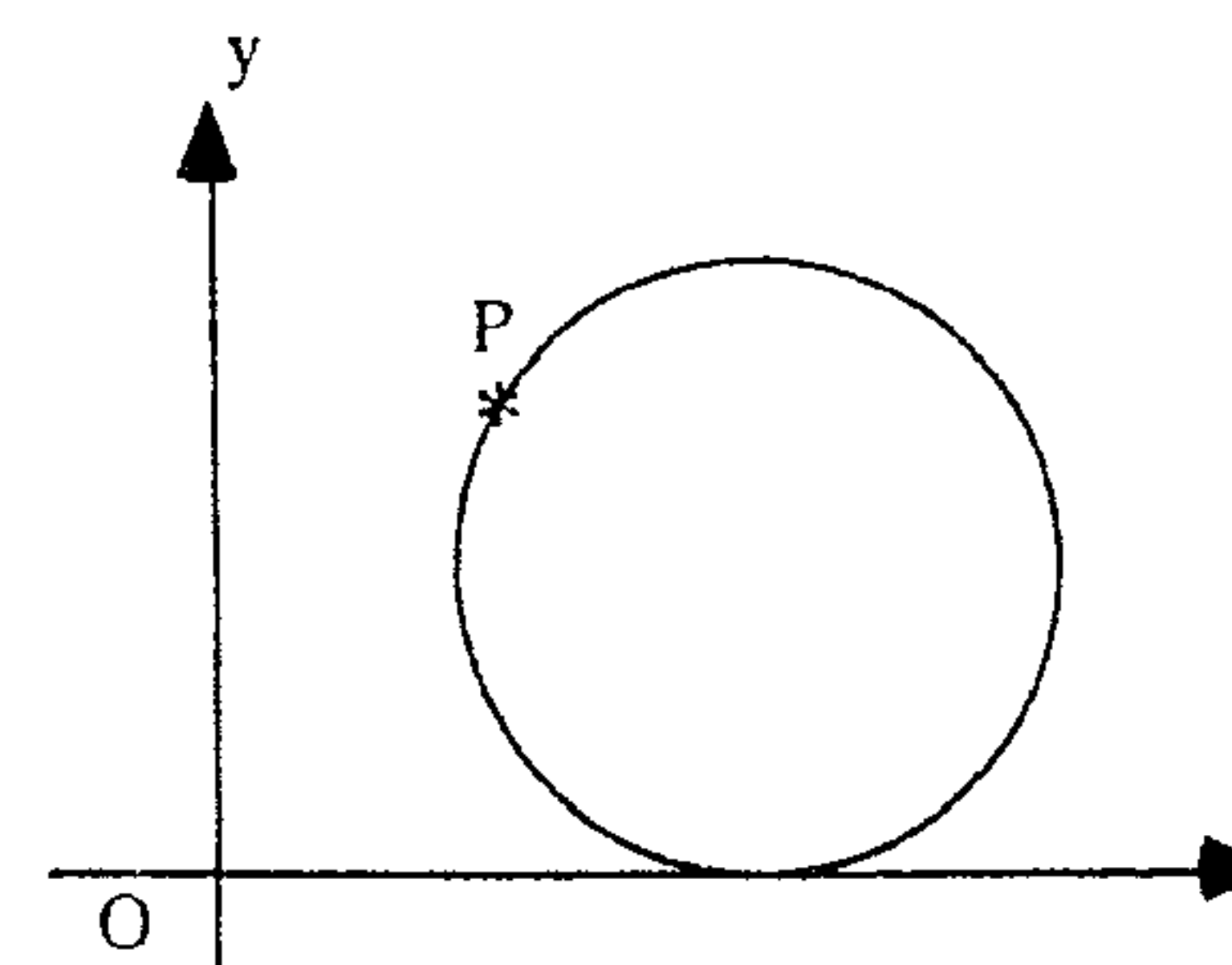
(b) A person borrows \$ 90,000 at an interest rate of $7\frac{1}{2}\%$ per annum compounded monthly. It is to be paid with equal monthly instalments over 15 years. Find:

(i) the value of each monthly repayment.

(ii) the amount remaining on the loan after the 120th payment.

(iii) how many repayments are needed if the monthly instalment is increased to \$ 1000 after the 120th payment.

Question 4 : START A NEW PAGE



The cycloid curve is described by the locus of a fixed point P on the circumference of a circle as the circle moves horizontally along the x-axis. The parametric equations of a cycloid are given by :

$$x(\theta) = \theta - \sin \theta \quad \text{and} \quad y(\theta) = 1 - \cos \theta.$$

(i) Copy and fill in the following table.

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
x					
y					

(ii) Show that the y-axis is tangent to the cycloid at $\theta = 0$.

(iii) On a set of x and y axes, graph from the above table the path of P for $0 \leq \theta \leq 2\pi$.

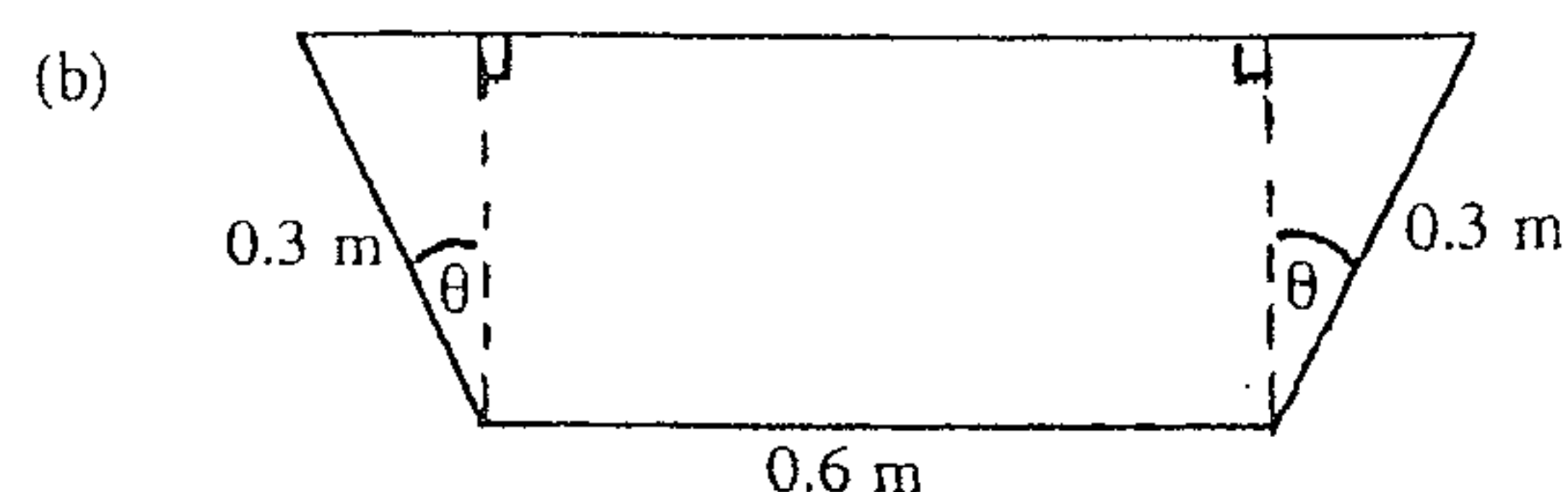
(iv) The area bounded by the path of P for $0 \leq \theta \leq 2\pi$ and the x-axis is given by :

$$\text{Area} = \int_0^{2\pi} (1 - \cos \theta)^2 d\theta$$

Evaluate the exact area.

Question 5: START A NEW PAGE

- (a) (i) Graph, on the same axes: $y = \sec x$ and $y = \tan x$ for $0 \leq x < \frac{\pi}{2}$.
- (ii) The area bounded by the positive y-axis, the curves $y = \sec x$ and $y = \tan x$, and the line $x = N$, where $0 < N < \frac{\pi}{2}$, is rotated about the x-axis. Find the volume of revolution in terms of N .
- (iii) Find the volume as N approaches $\frac{\pi}{2}$.

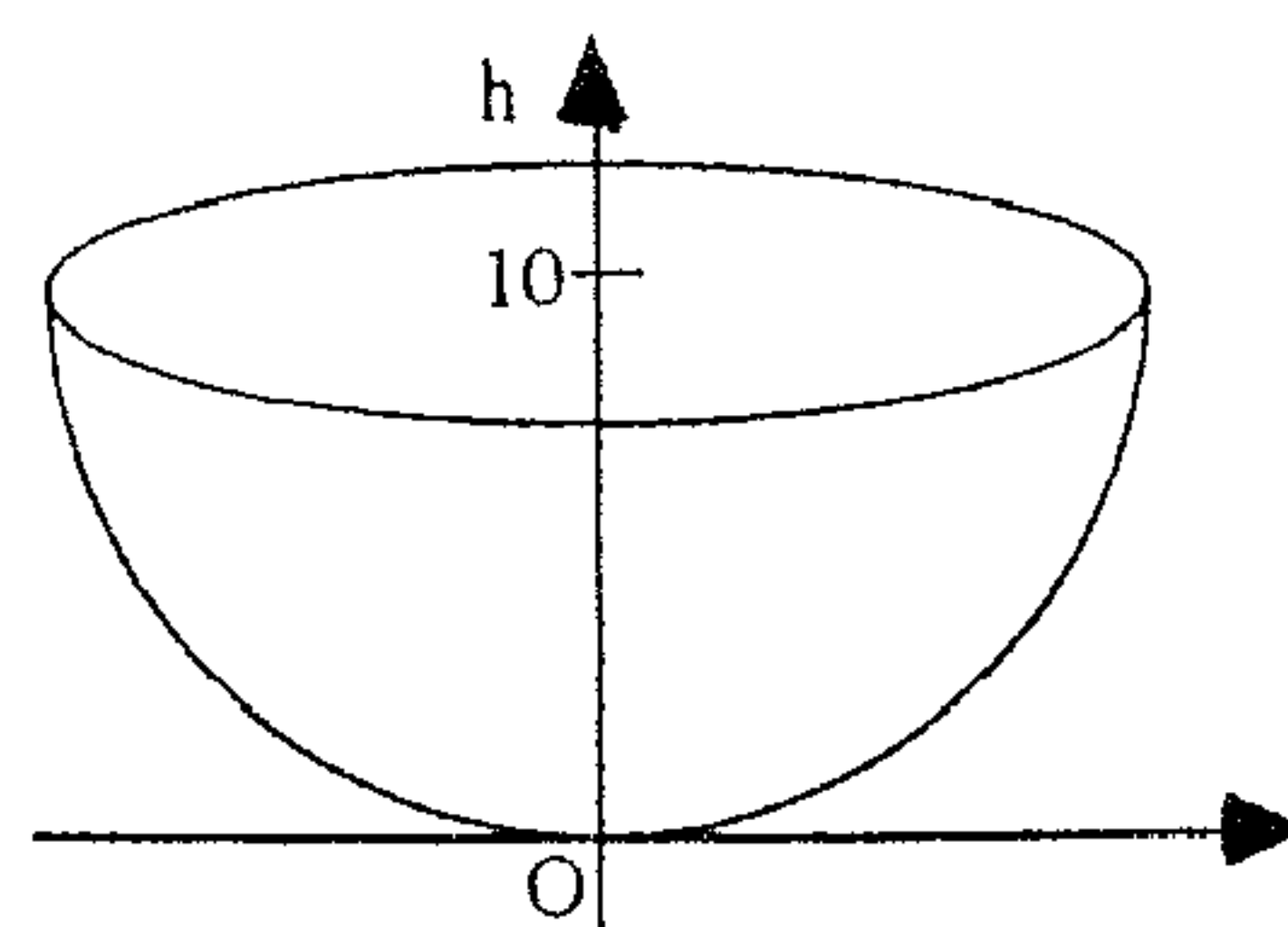


A farmer is to make a discharge chute with a trapezoidal cross-section as shown.

- (i) Show the cross-sectional area A is given by: $A = 0.09 \cos \theta (2 + \sin \theta) \text{ m}^2$
- (ii) Hence find the angle θ (to the nearest degree for $0 < \theta < 90^\circ$) which will maximise the cross-sectional area A .

Question 6: START A NEW PAGE

(a)



An empty hemi-spherical bowl 20 cm. in diameter rests on a flat surface. Water is poured into the bowl at a constant rate of $20 \text{ cm}^3 / \text{minute}$. After t minutes, the water level is h cm. above the base of the bowl. Find the rate at which the water level is rising when the deepest water in the bowl is 6 cm. [You may use $V = \frac{1}{3} \pi h^2 (30 - h)$]

- (b) (i) Draw a neat sketch of $y = f(x)$, where $f(x) = 2 \sin^{-1} x$.
- (ii) On the same axes, graph and **clearly label** $y = f^{-1}(x)$, the inverse of $y = f(x)$.
- (iii) Find the exact value of the area bounded by the curves $y = f(x)$ and $y = f^{-1}(x)$, and the line $x = 1$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

THIS IS THE END OF THE PAPER

$$(u) \sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6} \quad (1)$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin 4x}{2x} = \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \cdot 2 = 2 \quad (2)$$

$$(c) (i) \frac{d}{dx} 2 \tan x^3 = 6x^2 \sec^2(x^3) \quad (2)$$

$$(ii) \frac{d}{dx} x \sin^{-1} x = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$(d) y = \operatorname{cosec} x$$

$$y' = -\operatorname{cosec} x \cot x$$

$$x = \frac{7\pi}{6} \quad m = -\operatorname{cosec} \frac{7\pi}{6} \cot \frac{7\pi}{6}$$

$$= -(-2) \cdot \sqrt{3}$$

$$= 2\sqrt{3} \quad (2)$$

$$e) \frac{d}{dx} \cos^{-1}\left(\frac{1}{x}\right) = \frac{-1}{\sqrt{1-\frac{1}{x^2}}} \cdot \frac{-1}{x^2}$$

$$= \frac{1}{\sqrt{x^2-1}} \cdot \frac{1}{x^2}$$

$$= \frac{1}{x\sqrt{x^2-1}} \quad (3)$$

$$2 (a) \int_{-4}^4 \frac{6 dx}{x^2+16} = 6 \cdot \frac{1}{4} \left[\tan^{-1} \frac{x}{4} \right]_{-4}^4$$

$$= \frac{3}{2} \left\{ \tan^{-1} 1 - \tan^{-1}(-1) \right\}$$

$$= \frac{3}{2} \left\{ \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \right\}$$

$$= \frac{3\pi}{4} \quad (2)$$

$$b) \int \frac{2 dx}{\sqrt{3-4x^2}} = \int \frac{2 dx}{\sqrt{4\left(\frac{3}{4}-x^2\right)}}$$

$$= \sin^{-1} \frac{2x}{\sqrt{3}} + C \quad (2)$$

$$\int \frac{1}{\sqrt{9-x^2}} \left(4x(9-x^2) + \sqrt{9-x^2} \right)^{1/2} dx$$

$$= -4\sqrt{9-x^2} + 5 \sin^{-1} \frac{x}{3} + C \quad (3)$$

$$d(i) 143 \cos \theta - 24 \sin \theta = R \cos(\theta + \alpha)$$

$$= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$\therefore R \sin \alpha = 24 \quad R > 0 \quad 0 < \alpha < \frac{\pi}{2}$$

$$R \cos \alpha = 143$$

$$R = \sqrt{24^2 + 143^2}$$

$$= 145$$

$$\tan \alpha = \frac{24}{143}$$

$$\alpha = 0.1663$$

$$i. 143 \cos \theta - 24 \sin \theta = 145 \cos(\theta + 0.1663) \quad (2)$$

$$ii) 143 \cos \theta - 24 \sin \theta = 100$$

$$145 \cos(\theta + 0.1663) = 100$$

$$\cos(\theta + 0.1663) = \frac{100}{145}$$

$$\theta + 0.1663 = 0.8098, 5.4734 \quad (3)$$

$$\theta = 0.6435, 5.3071 \quad 0 < \theta < 2\pi$$

$$3 (a) S_n = 3^n + 3^{n-1} + \dots + 3^{-2n}$$

$$S_n = 3^{n-1} + 3^{n-2} + \dots + 3^{-2n} + 3^{-2n-1}$$

$$\therefore \frac{2}{3} S_n = 3^n - 3^{-2n-1}$$

$$S_n = \frac{1}{2} \left[3^{n+1} - 3^{-2n} \right] \quad (3)$$

Principal $P = \$90000$ Interest Rate = $7\frac{1}{2}\%$ pa
 $= 0.625\%$ per month

i) let R be repayment per month

$$\therefore \text{Amount owing after 1st Payment} = P(1 + 0.00625) - R$$

$$A_1 = 1.00625P - R$$

$$\text{Amount owing after 2nd Payment} = (1.00625P - R)1.00625 - R$$

$$A_2 = 1.00625^2 P - R[1 + 1.00625]$$

$$\text{Amount owing after 3rd Payment} = (1.00625^2 P - R[1 + 1.00625])1.00625 - R$$

$$A_3 = 1.00625^3 P - R[1 + 1.00625 + 1.00625^2]$$

$$\text{Amount owing after } n \text{ Payments} = 1.00625^n P - R[1 + 1.00625 + \dots + 1.00625^{n-1}]$$

$$A_n = 1.00625^n P - R \left[\frac{1.00625^n - 1}{0.00625} \right] \quad (1)$$

For 15 years $n = 180$ $A_{180} = 0$ $P = 90000$

$$\therefore R \left[\frac{1.00625^{180} - 1}{0.00625} \right] = 90000 \cdot 1.00625^{180}$$

$$R = \frac{90000 \cdot 1.00625^{180} \cdot 0.00625}{1.00625^{180} - 1}$$

Monthly Repayment = \$834.31 (4)

ii) $A_{120} = 90000 \cdot 1.00625^{120} - 834.31 \left(\frac{1.00625^{120} - 1}{0.00625} \right)$

Amount owing End 120th Payment = \$41636.75 (2)

iii) $R = 1000$ $P = 41636.75$, $A = 0$ by θ

$$1.00625^n \cdot 41636.75 = \frac{1000 (1.00625^n - 1)}{0.00625}$$

$$= 160000 \cdot 1.00625^n - 160000$$

$$1.00625^n = \frac{160000}{118363.25}$$

$$= 1.35$$

$$n = \frac{\ln 1.35}{\ln 1.00625} = 48.38 \Rightarrow 49 \text{ payment} \quad (3)$$

4(i)

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
x	0	$\frac{\pi}{2} - 1$	π	$\frac{3\pi}{2} + 1$	2π
y	0	1	2	1	0

(3)

(ii)

$$y = 1 - \cos \theta$$

$$x = \theta - \sin \theta$$

$$\frac{dy}{d\theta} = \sin \theta$$

$$\frac{dx}{d\theta} = 1 - \cos \theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta} \quad (1)$$

At $\theta = 0$

$$\frac{dy}{dx} = \lim_{\theta \rightarrow 0} \left(\frac{\sin \theta}{1 - \cos \theta} \cdot \frac{1 + \cos \theta}{1 + \cos \theta} \right)$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin \theta (1 + \cos \theta)}{1 - \cos^2 \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin \theta (1 + \cos \theta)}{\sin^2 \theta}$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{1 + \cos \theta}{\sin \theta} \right)$$

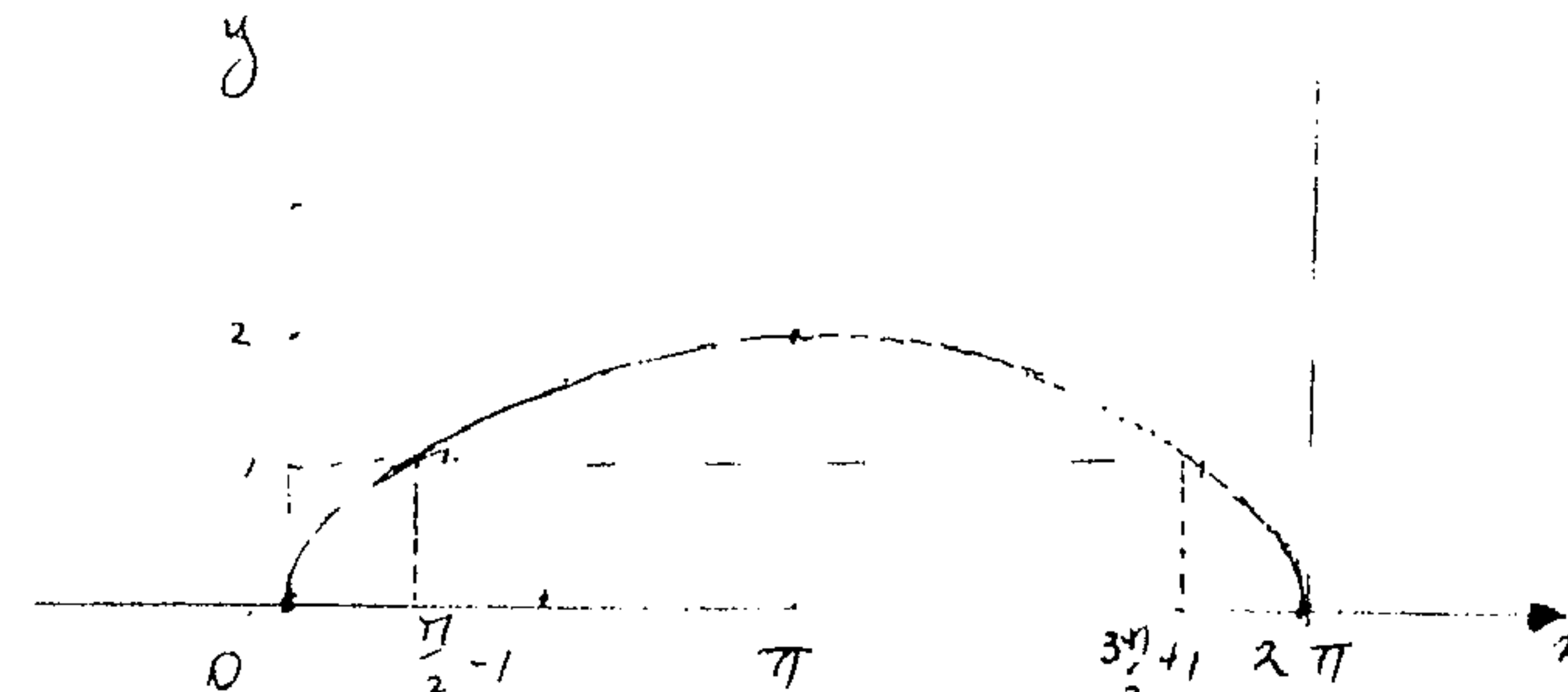
$$= \infty$$

iv) gradient is vertical when $\theta = 0$

v) Equation Tangent is $x = 0$

$\therefore y$ axis is a tangent to the cycloid

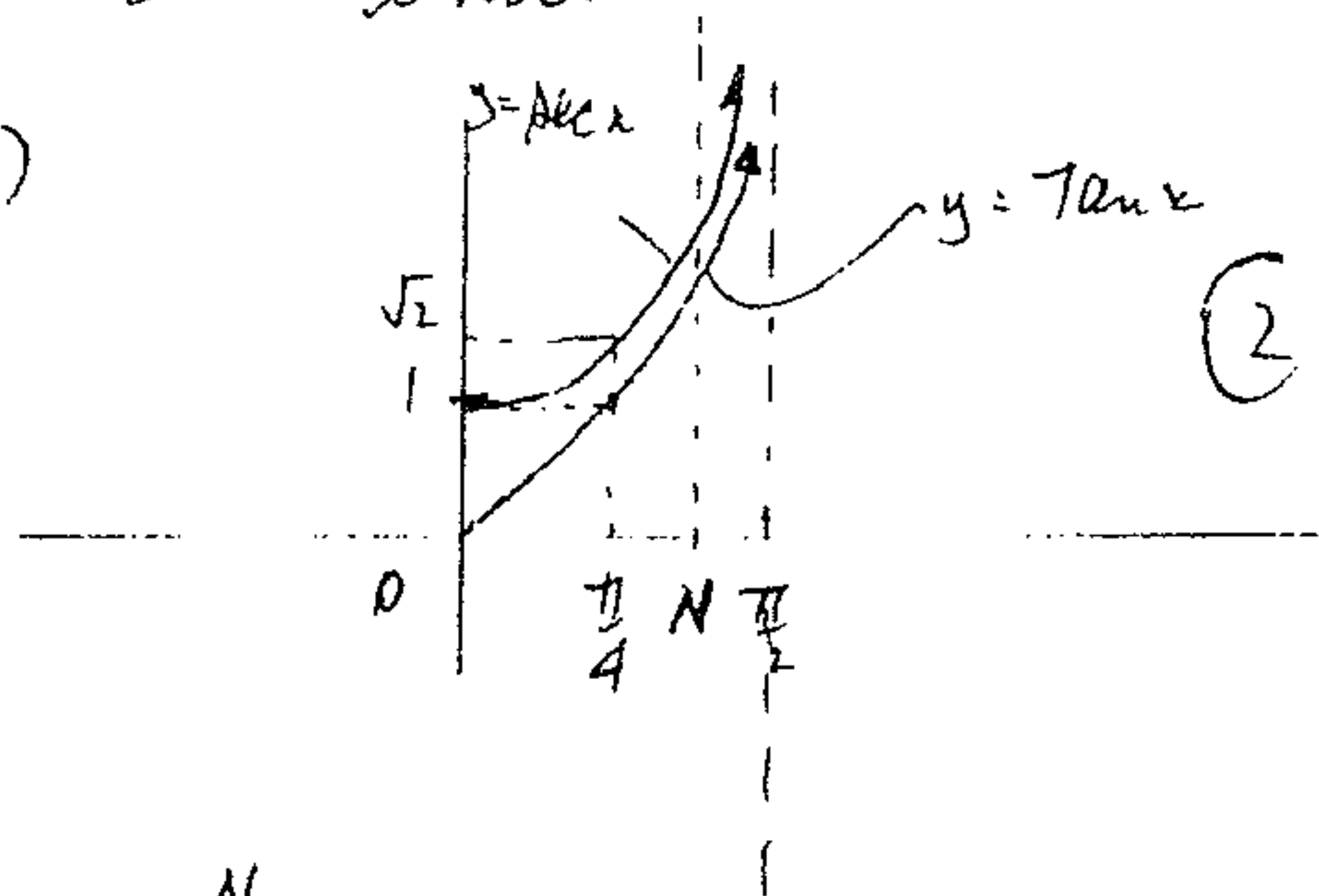
iii)



(2)

$$\begin{aligned}
 10) \quad A &= \int_0^{2\pi} (1 - \cos \theta)^2 d\theta \\
 &= \int_0^{2\pi} (1 - 2\cos \theta + \cos^2 \theta) d\theta \\
 &= \int_0^{2\pi} 1 - 2\cos \theta + \frac{\cos 2\theta + 1}{2} d\theta \\
 &= \int_0^{2\pi} \frac{3}{2} - 2\cos \theta + \frac{1}{2} \cos 2\theta d\theta \\
 &= \left[\frac{3}{2} \theta - 2 \sin \theta + \frac{\sin 2\theta}{4} \right]_0^{2\pi} \\
 &= 3\pi - 0 + 0 - 0 \\
 &= 3\pi \text{ units}^2
 \end{aligned}$$

5 (a) (i)



(ii)

$$\begin{aligned}
 V &= \pi \int_0^N (y_1^2 - y_2^2) dx \\
 &= \pi \int_0^N \sec^2 x - \tan^2 x dx \\
 &= \pi \int_0^N 1 dx \\
 &= \pi [x]_0^N \\
 &= \pi N \text{ units}^3
 \end{aligned}$$

$$\text{(iii)} \quad \lim_{N \rightarrow \frac{\pi}{2}} V = \frac{\pi^2}{2} \text{ units}^3$$

$$\begin{aligned}
 6) \quad \text{Area Trapezium} &= \frac{1}{2} (x+y) \\
 &= \frac{1}{2} \cdot 0.3 \cos \theta [0.6 + 0.6 + 2 \times 0.3 \sin \theta] \\
 7) \quad A &= 0.09 \cos \theta [2 + \sin \theta] \\
 A' &= 0.09 \{ -\sin \theta (2 + \sin \theta) + \cos^2 \theta \} \\
 &= 0.09 \{ \cos^2 \theta - \sin^2 \theta - 2 \sin \theta \} \\
 &= 0.09 \{ \cos 2\theta - 2 \sin \theta \} \\
 A'' &= 0.09 \{ -2 \sin 2\theta - 2 \cos \theta \}
 \end{aligned}$$

For maximum area

$$A' = 0$$

$$\cos^2 \theta - \sin^2 \theta - 2 \sin \theta = 0$$

$$1 - 2 \sin^2 \theta - 2 \sin \theta = 0$$

$$2 \sin^2 \theta + 2 \sin \theta - 1 = 0$$

$$\sin \theta = \frac{-2 \pm \sqrt{4+8}}{4}$$

$$= \frac{\sqrt{3}-1}{4} \text{ only } \sin \theta > 0 \quad 0 < \theta < 90^\circ$$

$$\theta = 0.37 \text{ rad}$$

$$\theta = 21^\circ \text{ (nearest degree)}$$

we will test A'' for sign concavity, since A, A', A'' are continuous in domain $0 < x < \frac{\pi}{2}$

$$A''(0.37) = -0.288 < 0$$

$\therefore$ There is a relative maximum at $\theta = 21^\circ$, but since there is only one turning point in domain $0 < \theta < \frac{\pi}{2}$ there is an absolute maximum cross-sectional area when $\theta = 21^\circ$.

$$b(a) \quad V = \frac{1}{3} \pi h^2 (30-h)$$

$$V = \frac{\pi}{3} (30h^2 - h^3)$$

$$\frac{dV}{dh} = \frac{\pi}{3} (60h - 3h^2)$$

$$= \pi (20h - h^2)$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$\frac{dV}{dt} = \pi (20h - h^2) \frac{dh}{dt}$$

$$h = 6 \text{ cm} \quad \frac{dV}{dt} = 20 \text{ cm}^3/\text{minute}$$

$$20 = \pi (20 \times 6 - 6^2) \cdot \frac{dh}{dt}$$

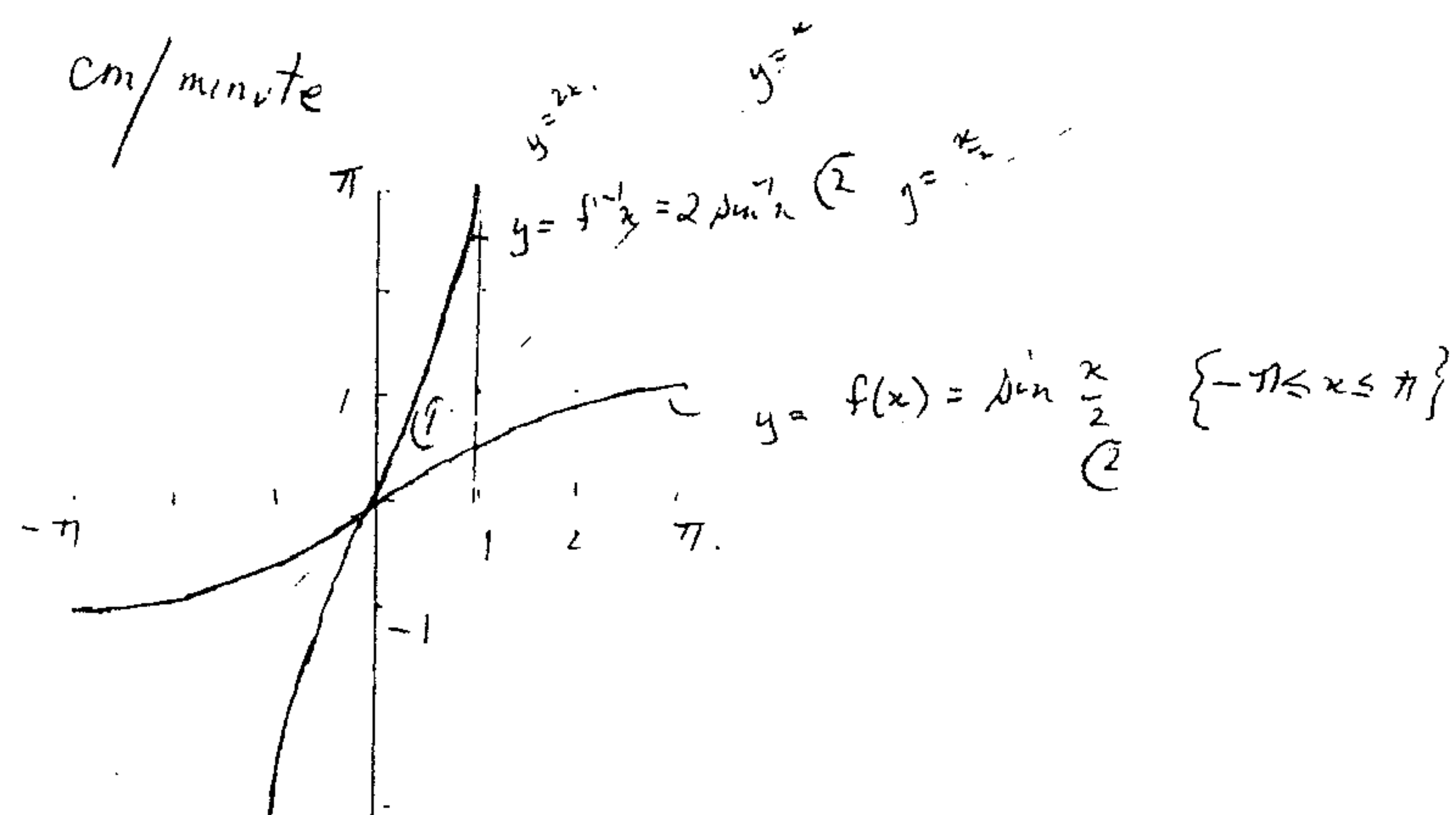
$$\frac{dh}{dt} = \frac{20}{84\pi}$$

(3)

rate
rise

$$= \frac{5}{21\pi} \text{ cm/minute}$$

b(i)



$$A = \int_0^1 2 \sin^{-1} x \, dx - \int_0^1 \sin \frac{x}{2} \, dx$$

$$= \pi - \int_0^1 \sin \frac{x}{2} \, dx - \int_0^1 \sin \frac{x}{2} \, dx$$

$$= \pi + \left[2 \cos \frac{x}{2} \right]_0^1 + \left[2 \cos \frac{x}{2} \right]_0^1$$

$$= \pi + 0 - 2 + 2 \cos \frac{1}{2} - 2$$

$$= \pi + 2 \cos \frac{1}{2} - 4 \text{ units}^2$$

(4)